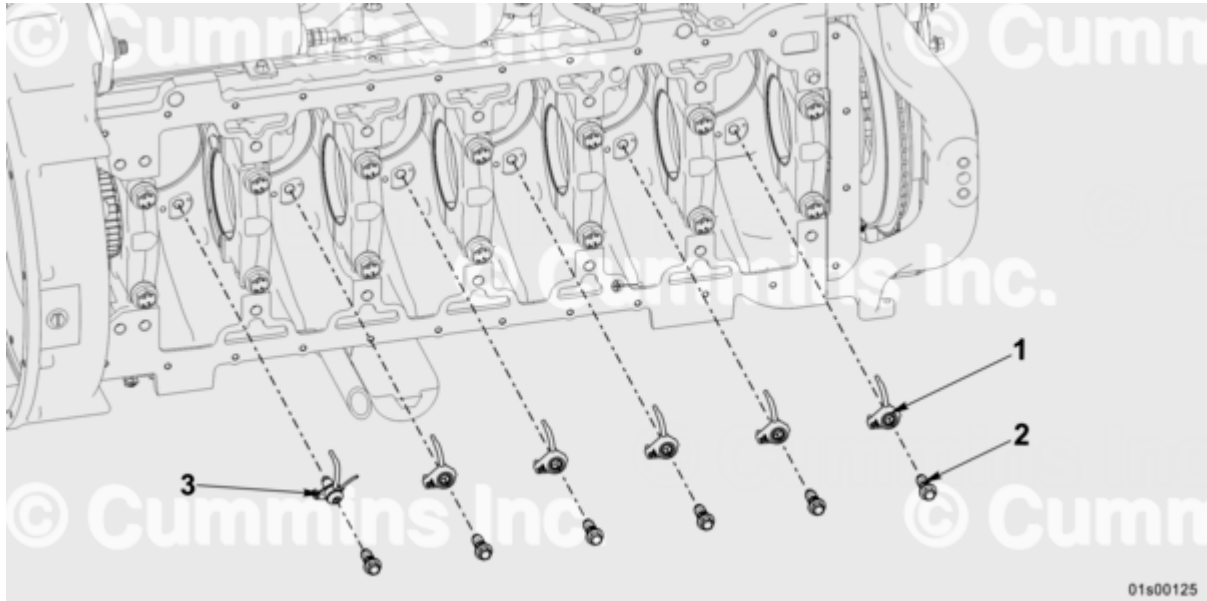


Trainings ▾

Exploded View



Piston Cooling Nozzle Exploded View

1. Piston cooling nozzle - quantity 5
2. Capscrew - quantity 6
3. Piston cooling nozzle - quantity 1

Select Service Tools

Recommended Cummins® Service Tools

- No recommended Cummins® service tools required

Additional Service Items

- No additional service items required.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

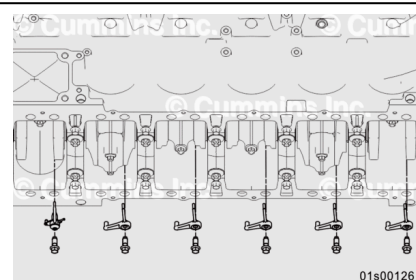
Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the lubricating oil system. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/1016/1016-007-037.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/1016/1016-007-025.html)
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1.
(/qs3/pubsys2/xml/en/procedures/1016/1016-001-089.html)

Remove

⚠ CAUTION ⚠

Do not drop or mishandle the piston cooling nozzles during removal. Damage can occur to the piston cooling nozzle. If the piston cooling nozzle is damaged, engine damage can occur.

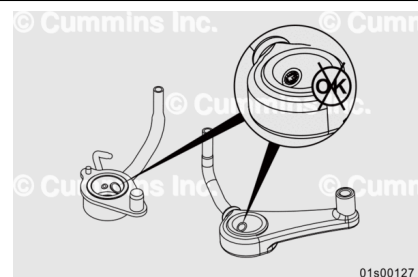


The crankshaft **must** be rotated to allow access to remove all of the nozzles. Refer to Procedure 000-017 in Section 0.
(/qs3/pubsys2/xml/en/procedures/1016/1016-000-017.html)

Remove the piston cooling nozzle capscrews and nozzles.

Clean and Inspect for Reuse

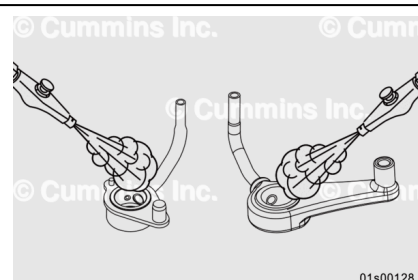
Inspect the piston cooling nozzles for damage. If any damage is found, replace the piston cooling nozzle.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean with compressed air. Run air through the piston cooling nozzle and banjo fitting.



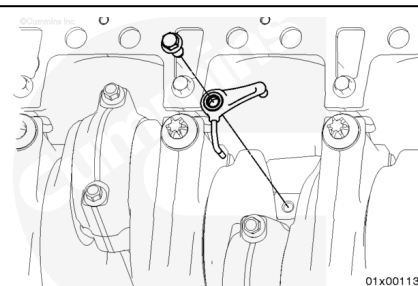
Install

The crankshaft **must** be rotated to allow access to install all of the cooling nozzles. Refer to Procedure 000-017 in Section 0. (</qs3/pubsys2/xml/en/procedures/1016/1016-000-017.html>)

Align and install the piston cooling nozzle and mounting capscrew.

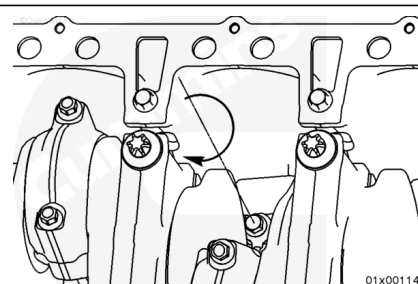
Make sure the locating pin is inserted into the block.

Make sure the piston cooling nozzle is pointing to the cool side of the block.



Tighten the piston cooling nozzle mounting capscrews.

Torque Value: 65 n·m [48 ft·lb]



Finishing Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1.
(/qs3/pubsys2/xml/en/procedures/1016/1016-001-089.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/1016/1016-007-025.html)
- Fill the lubricating oil system. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/1016/1016-007-037.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.